

Qatar-1b

R-1304

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_241123 · created 2026-07-31T17:36:10+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460637.6033 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.034
Transit depth (Rp/Rs)^2	3.62 +/- 1.29 [%]
Orbital Inclination (inc)	82.09 +/- 1.75
Airmass coefficient 1 (a1)	0.977 +/- 0.013
Airmass coefficient 2 (a2)	0.016 +/- 0.041
Scatter in the residuals of the lightcurve fit is	1.35 %
Best Comparison Star	#1 - [324, 458]
Optimal Aperture	3.38
Optimal Annulus	12.24
Transit Duration (day)	0.044 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.19 $\pm$ 0.034 vs 0.1463 (deviation 1.29 σ)
Duration fit vs published	0.044 d vs 0.0692 d (deviation 1.94 σ)
Mid-time O–C	6.3 min
Depth SNR	5.6
Residual scatter	1.35 %

Light curve

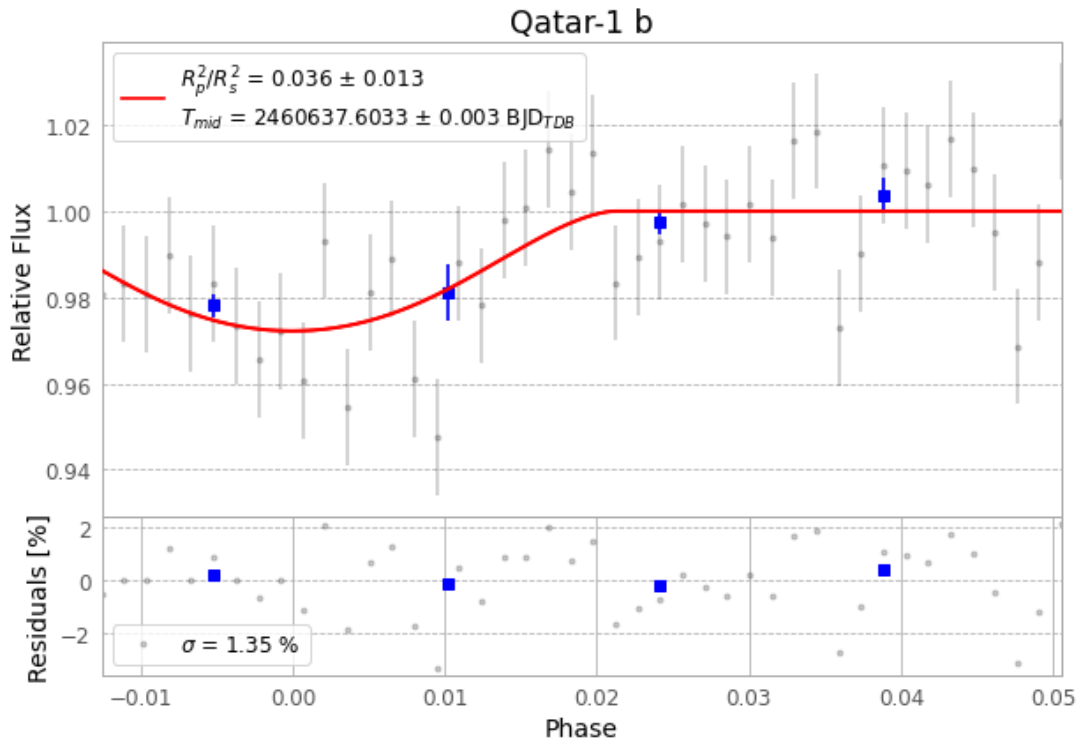


Figure 1 — FinalLightCurve_Qatar-1 b_2024-11-22.png (SHA-256 9f1db0e5fea568c5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [362, 286], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4ff774ec4b6b958f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

44 FITS frames (29 MB), Qatar-1241123020015.FITS → Qatar-1241123040915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-241123.log (a18e10c4c9b6...), FinalLightCurve_Qatar-1 b_2024-11-22.png (9f1db0e5fea5...), FinalLightCurve_Qatar-1 b_2024-11-22.csv (aa5fcb36abd1...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 17:36Z.